REFERENCE APPLICATION · CAPABILITY BRIEF

Ask the Wizard anything about a pinball machine — every answer cites the manufacturer's own manual, or refuses rather than guess.

PinballWizard is a live, customer-facing reference application. The pinball domain is the vehicle; the engineering is the point — proof that EarlyBird can architect, ship, and operate an enterprise-class AI system on Azure, end to end.

PREPARED BY	FOR	LIVE AT	DATE
EarlyBird Solutions	Prospective clients	pinwiz.ai	July 1, 2026

THE SYSTEM

A polite scraper feeding a source-citing RAG Wizard

Ten polite scrapers crawl eight pinball manufacturers plus the canonical OPDB catalog and persist to Cosmos with full provenance. An event-driven pipeline turns those documents into a searchable corpus, and a tool-using agent answers questions over it — grounded, cited, and honest about its limits.

> “
>
> **Every answer ends with a clickable citation back to the original manufacturer source — or it refuses, rather than fabricate.**

THE DIFFERENTIATOR

Provenance, not plausibility

The failure mode of a generic RAG demo is a confident, plausible-sounding answer with no way to check it. PinballWizard is built the other way around: the citation chain is the product.

GENERIC RAG

Retrieves some text, generates a fluent answer, and can hallucinate a source — or invent one — with nothing to verify against.

PINBALLWIZARD

Every claim resolves through a deterministic `document_id` back to the exact page and file it came from; below a confidence threshold it refuses instead of guessing.

CAPABILITIES

What a prospect can verify in the repository

Each of these is inspectable in code, tests, and Architecture Decision Records — not just described in a slide.

- ✅ **Cloud-native architecture** — Azure Container Apps, Cosmos DB, AI Search, and Azure OpenAI / Microsoft Foundry, orchestrated locally with .NET Aspire mirroring production.
- ✅ **AI engineering** — event-driven RAG (Change Feed → chunk → embed → index), a four-agent Foundry surface with function tools, two-stage re-ranking, and confidence-threshold refusal.
- ✅ **Real-time streaming UI** — a Blazor front end streaming answers over Server-Sent Events with source-citation strips and honest refusal panels.

- **Engineering discipline** — Clean Architecture enforced by fitness tests, 47 ADRs for non-obvious decisions, a behavior-asserting test suite, and a zero-warning build.
- **Infrastructure-as-code & operability** — two-tier Bicep, OpenTelemetry throughout, health checks, metric alerts, and operational runbooks.
- **Polite integration** — `robots.txt` honored unconditionally, machine-consumer metadata preferred over DOM scraping, identifying user-agents, throttled by construction.

THE DELIVERY MODEL

AI-authored, human-governed

Nearly all of the code is written by AI — and held to an enterprise bar by a process where a human owns intent, judgment, and the merge button. Nothing merges on faith.

AI · AUTHORS

Speed and consistency

Spec → plan → test-first implementation, applied consistently across the codebase.

- Behavior-asserting tests as documentation of intent
- Standards and invariants bound every change

HUMAN · GOVERNS

Judgment and accountability

Every change passes the same gates regardless of who typed it.

- First-party review, CI gates, independent CodeQL
- Whole-branch senior review, then a human merge

AT A GLANCE

The envelope

SOURCES

8 + OPDB

Manufacturers scraped politely, plus the canonical machine catalog.

AGENTS

Four

Wizard, Valuation, Rules, and Repair — a tool-using surface, not single-pass RAG.

DECISIONS

47 ADRs

Every non-obvious choice recorded, traceable, and append-only.

COST CAP

$300–$400/mo

Steady-state budget with cost-per-feature attribution — cheap to evaluate.

TESTS

2,900+

Behavior-asserting, across seven per-layer projects; zero-warning build.

STATUS

Live

Running on Azure at **pinwiz.ai**; the repository is the showcase artifact.